Fiscal Year:	FY 2016	**Task Last Updated:**	FY 05/02/2016
PI Name:	Chang, Polly Y. Ph.D.		
Project Title:	Particle Radiation-Induced Dysregulation of Protein Homeostasis in the Brain		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	polly.chang@sri.com	**Fax:**	FY 650-859-2889
PI Organization Type:	NON-PROFIT	**Phone:**	650-859-2549
Organization Name:	SRI International		
PI Address 1:	Biosciences, PN 187		
PI Address 2:	333 Ravenswood Avenue		
PI Web Page:			
City:	Menlo Park	**State:**	CA
Zip Code:	94025-3453	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
Start Date:	05/01/2014	**End Date:**	04/30/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Change in period of performance (original period of performance was 2/26/2014-2/25/2018) per PI and NSSC information (Ed., 12/9/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lin, Hua (SRI International) Shaler, Thomas (SRI International)		
Grant/Contract No.:	NNX14AE69G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The space atmosphere is comprised of wide spectrum of highly energetic and charged particle radiation. Exposure to this type of radiation in the galacto-cosmic environment or unplanned solar particle events (SPE) may pose a potential health hazard to humans in long-term manned space explorations. In this project, we aim to conduct in-depth investigations of those cellular pathways that regulate the removal of damaged proteins, repair of DNA damage, and response to conditions of oxidative stress through protein modifications in the brain. Protein modification by ubiquitin is a highly regulated process that is catalyzed by specific enzyme families that are capable of modifying the correct protein in the correct cellular location at the correct time in order to control important processes such as the maintenance of genomic stability, cell cycle progression, removal of damaged or misfolded proteins, impaired autophagy, apoptosis, and gene transcription. Depending on the radiation dose and beam quality, particle radiation may cause significant damage to proteins and oxidative stress in the cells in the central nervous system, leading to dysfunction of the cellular ubiquitin-related pathways, an increase in the levels of extracellular ubiquitin, and persistent subsequent downstream effects on regulation of stress-associated signaling pathways in many different cell types within the central nervous system (CNS), precipitating neurodegenerative disease. We hypothesize that quantitative changes in cellular ubiquitin pools occur when neuronal cells are exposed to particle radiation and aim to study the impact of protein degradation after particle radiation using both in vitro and in vivo model systems. Information gathered from this research will aid in reducing the uncertainties associated with predicting risks both in the estimation of risk to astronaut health and the design of appropriate countermeasures to ameliorate any potential negative effects in space flights.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We expect that results associated with the regulation of protein processing and degradation in brain tissues obtained from our studies will be useful to understand molecular mechanisms of action in normal neuronal cell responses in particle radiation clinical therapy scenarios.
Task Progress:	We are using an in vitro/in vivo approach to assess changes in primary mouse and human neuronal cells and animal model. Primary mouse neuronal stem cells, representing cells from different regions of the brain and human neural stem cells were used in these studies. C57Bl6 mice were used as the animal model. Cells or animals were irradiated with either 250 MeV proton or 1 GeV/n Fe ions. Cells lysates and brain tissues of the irradiated and sham–treated animals were harvested at different times post irradiation. Quantitative levels of ubiquitin pools and relative levels for global proteome changes were measured. We observed that particle radiation quality appeared to be important for protein ubiquitination in certain brain cells/tissues. Proteins harvested from hippocampus tissues at 11 wks after a single dose of Fe-ions showed significant accumulation of polyubiquitin (pUb)-linked chains relative to the controls. On the other hand, there was no significant difference in pUb chain linkages seen in the cortical tissues from the same Fe-irradiated animals, showing brain-region specificity. In contrast, no significant difference in the levels of pUb chains from brain tissues from animals after proton-irradiation was observed. Interestingly, quantitative results for pUb-proteins in the different murine neuronal cultures also showed no significant differences when compared to the controls, supporting the results of the in vivo findings for the proton beam. In addition to the pUb chains, about 1000 proteins were profiled and several proteins known to be critically important in the functional integrity of the neuronal system were found to be upregulated. In conclusion, particle fluences play a role in targeted protein degradation in neuronal tissues and that our in vitro results support results obtained from the animal studies. Our results show that long-term changes in protein turnover pathways related to the ubiquitin proteasome system can occur in specific brain regions as a result of exposure to particle irradiation.
Bibliography Type:	Description: (Last Updated: 04/26/2023)
Abstracts for Journals and Proceedings	Shaler T, Lin H, Chen S, Bakke J, Du N, Chang P. "Particle Radiation-induced Dysregulation of Protein Homeostasis in the Brain." Presented at the 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016
Abstracts for Journals and Proceedings	Shaler TA, Lin H, Chang PY. "Particle radiation induced dysregulation of protein homeostasis in the brain." Presented at the 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015
Abstracts for Journals and Proceedings	Shaler T, Lin H, Bakke J, Du N, Tilton A, Chen S, Chang P. "Initial Investigations of possible particle radiation-induced dysregulation of protein homeostasis in the brain" Presented at the 61st Annual Meeting of the Radiation Research Society, Weston, FL, September 19-22, 2015. 61st Annual Meeting of the Radiation Research Society, Weston, FL, September 19-22, 2015. , Sep-2015